

Linear Algebra Practice

1. Use the following code to read in data and store the results in `Credit`

```
Credit <- read.csv("http://statlearning.com/s/Credit.csv")
```

2. Create a linear model object named `mod_num` by regressing `Balance` onto `Income`, `Limit`, `Rating`, `Cards` and `Age`.
3. Create the model matrix and store the results in `X`.
4. Create the matrix $X'X$ and store the result in `XTX`.
5. Find the inverse of $(X'X)^{-1}$ and store the result in `XTXI`.
6. Verify that the answer you computed in 5. is the same thing as `summary(mod_num)$cov.unscaled`.
7. Use the normal equations to find the coefficients for `mod_num`. Verify that your estimates computed with the normal equations are the same as `coef(mod_num)`.
8. What is the MSE for `mod_num`.
9. Compute the variance covariance matrix for your estimated coefficients. Store the values in an object named `var_cov_b`.
10. Compute the standard errors for your estimated coefficients. Verify that your answers are the same as `summary(mod_num)$coef[, 2]`.
11. Compute the t values using your answers from 7. and 10. Verify that the values are the same as `summary(mod_num)$coef[, 3]`.
12. Compute the p-values for testing $H_0 : \beta_i = 0$ versus $H_0 : \beta_i \neq 0$ for $i = 0, \dots, 5$.